

Figure 3: Outlier Recall at top L% from the ranked list of outliers for all the datasets. ONE, though it is an unsupervised algorithm, outperforms all the baseline algorithms in most of the cases. SEANO uses 20% labeled data for training.

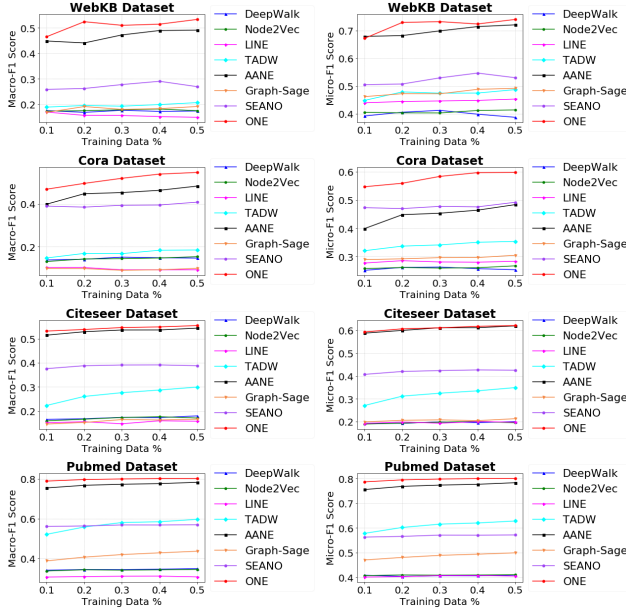


Figure 4: Performance of different embedding algorithms for Classification with Random Forest

tering assignments discovered by some algorithm, and $\mathcal{P}$ is a permutation of the set of labels. We assume 1 to be a logical operator which returns 1 when the argument is true, otherwise returns 0. Clustering performance is shown and explained in Fig. 5. It can be observed that except for ONE, all the conventional embedding algorithms fail in the presence of outliers. Our proposed unsupervised algorithm is able to outperform or remain close to SEANO, though SEANO is semi supervised in nature, on all the datasets.

6 Discussion and Future Work

We propose ONE, an unsupervised attributed network embedding approach that jointly learns and minimizes the effect of outliers in the network. We derive the details of the algorithm to optimize the associated cost function of ONE. Through experiments on seeded real world datasets, we show the superiority of ONE for outlier detection and other downstream network mining applications.

There are different ways to extend the proposed approach in the future. One interesting direction is to parallelize the algorithm and check its performance on real world large at-

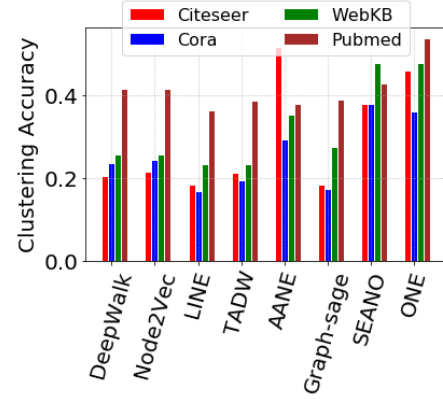


Figure 5: Clustering Accuracy of KMeans++ on the embeddings generated by different algorithms. ONE is always close to the best of the baseline algorithms. AANE works best for Citeseer. Though SEANO uses 20% labeled data as the extra supervision to generate the embeddings, its accuracy is always close (or less) to ONE which is completely unsupervised in nature.

tributed networks. Most of the networks are very dynamic now-a-days. Even outliers also evolve over time. So bringing additional constraints in our framework to capture the dynamic temporal behavior of the outliers and other nodes of the network would also be interesting. As mentioned in Section 4.9, ONE converges very fast on real datasets. But updating most of the variables in this framework takes $O(N)$ time, which leads to $O(N^2)$ runtime for ONE without any parallel processing. So, one can come up with some intelligent sampling or greedy method, perhaps based on random walks, to replace the expensive sums in various update rules.

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